



Gebze Technical University Faculty of Engineering

Group 8

Library Management System

Project Final Report

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## 2 Problem Definition:

Many institutions need systems where they can enter their data and access this data easily when requested. Especially libraries need these systems. In the absence of these systems, that is, when the data is kept manually, many problems are encountered by the library staff and users. These problems are:

- Information of libraries belonging to an institution,
- The information of the staff and in which library they work,
- Which books and magazines are in the libraries, the number of these books
- Location of books and magazines in the library
- Information of library users
- Information on which books users have delivered and which books they have not delivered
- Keeping information about books, for example; author of the book, number of pages, publishing house
- Keeping information about authors and publishers with books in the library
- Adding new library, staff and users to the system
- Updating existing libraries, staff and users information
- Filtering method suitable for the information sought in the system - Safe storage of data
- An easy interface for staff and users

To answer these problems, a system with different user profiles is needed. Thanks to this system, data can be stored securely and data loss is minimized. New data can be easily entered and existing data can be easily updated. The authorized persons can easily access the data from anywhere. Staff and users minimize time loss and workload. Makes it easier to track and access library materials such as books and magazines. By handling many tasks in the form of automation, it ensures that the margin of error is minimized. The system that provides these benefits and is recommended to respond to problems is the Library Management System.

## 3 Users of the System:

Library Management System consist 4 type of users.

### 3.1 Administrators:

Administrators are responsible for add new libraries to system or remove libraries from the system, add new library manager to system or remove library managers from system and see or edit library managers informations.

### 3.2 Library Manager:

Library manager are responsible for only their library. They add new Librarian to library or remove librarians, see-edit librarian's informations and search librarian, They update library informations like library librarians, library adress etc.

### 3.3 Librarian:

Librarians are responsible for only their library. They add new reader or delete reader, see reader's information or update reader's information, search reader, book, magazine, author, add new books magazines, update books-magazines and they update borrowed book list like, when the book has taken, when the book has given back and who has borrowed book.

### 3.4 Reader:

Readers can borrow books from only the library they are a member and they can loan magazines. They can search a books and magazines for it's category, author, name, publisher and rate. They can look for is the book, magazine available for now or has taken already and They can look place of the book in library which they want to borrow. They can rate a book

## 4 Requirements in Details:

### 4.1 Functional Requirements

The system will be used for administrator, library manager, librarian and reader. All users will enter the system with their user ID and passwords.

Administrators,

- Must add/remove library
- Must edit library

Library manager,

- Must add/remove Librarian
- Must edit Librarian
- Must see/search Librarian

- Must edit library

Librarian,

- Must add/remove Reader
- Must edit Reader
- Must add/remove book
- Must edit book
- Must add/remove loan book
- Must edit loan book
- Must add/remove author
- Must edit author
- Must add/remove category
- Must edit category
- Must add/remove publisher
- Must edit publisher
- Must add/remove magazine
- Must edit magazine
- Must search/see Reader
- Must search/see book
- Must search/see loan book and magazine

Reader,

- Must see loan books and magazines
- Must search/see book by book name
- Must search/see book by category
- Must search/see book by author
- Must search/see book by publisher
- Must search/see magazine by magazine name
- Must search/see magazine by category
- Must search/see magazine by author
- Must search/see magazine by Publisher

## **4.2 Non- Functional Requirements**

### **4.2.1 Usability Requirements**

The system is useful because users will perform their operations with an interactive Menu, they can select the option they want from the menu and proceed and username/password mechanism should be useful for user.

### **4.2.2 Security Requirements**

All passwords must be hard to guess and ideally require upper/lower case letters and special symbols to ensure high security also only admins can access this information.

#### **4.2.3 Performance Requirements**

The data in the system will be dynamic and this data will expand as users log in. At the same time, the algorithms we use will improve the performance.

#### **4.2.4 Space Requirements**

The data in the system will be dynamic and user information will be in this data such as book name, author, member information, employee information etc. Because of these, an expandable space is required.

#### **4.2.5 Operational Requirements**

The system must be in communication with the users in order to perform the necessary operations. User data and interaction is one of the most basic requirements in this system.

#### **4.2.6 Environmental Requirements**

The program is designed for the computer and will be written in a way to run on a computer terminal.

#### **4.2.7 Learnability Requirements**

When users see the interface, they can understand and implement the main actions.

#### **4.2.8 Accounting Requirements**

The system must have admin. Admin can confirm librarian, reader and library manager to register in the system. The operations to be performed by these accounts are determined by the administrators.

#### **4.2.9 Development Requirements**

This system will be developed by many developers at the same time, and the development environment must be suitable for it.

## 5 Use of Collections:

It had added due to feedback

### 5.1 List:

Reasons for using the list: We can keep more than one collection in the list reference (such as Linked list, Array List). It has an easy-to-use and well-known interface. It is generally used where it does not need to be sorted and where there is not much manipulation in the collection.

- Materials were also kept in the list for the author class. Here we used it to hold books or magazine written by the author.
- Materials were also kept in the list for the publisher class. Here we used it to hold books or magazine published by the publisher.
- Materials were also kept in the list for the user class. In this way, it will be possible to control which materials the users have used before.

### 5.2 Stack:

The Stack structure is used when the last added item will be used first.

- We used this data structure to hold the votes in the material class. Thus, the last rating in that book or journal will appear first.

### 5.3 Priority Queue:

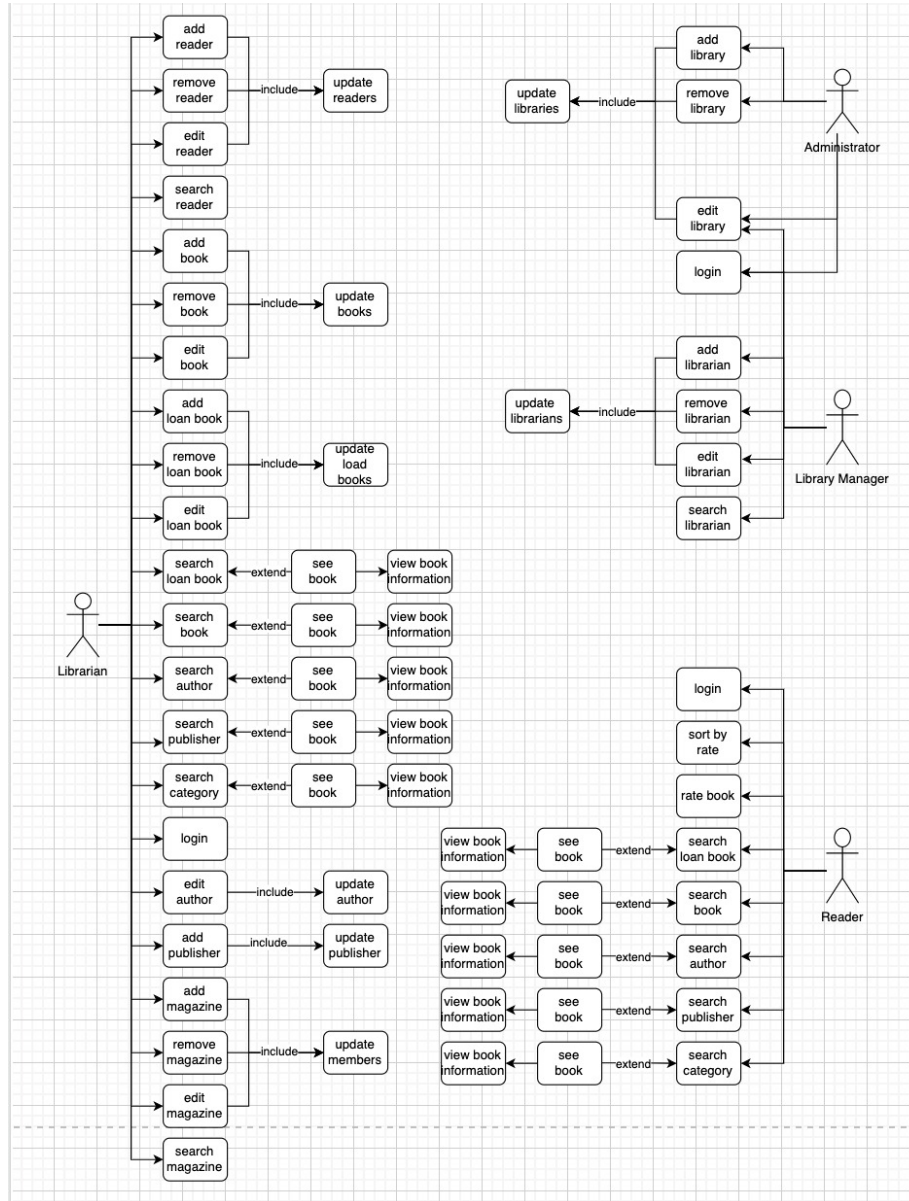
We have kept the books and magazines as a priority queue to keep them sorted by the height of the votes.

### 5.4 Binary Search Tree:

It is used in places where there is a need for searching.

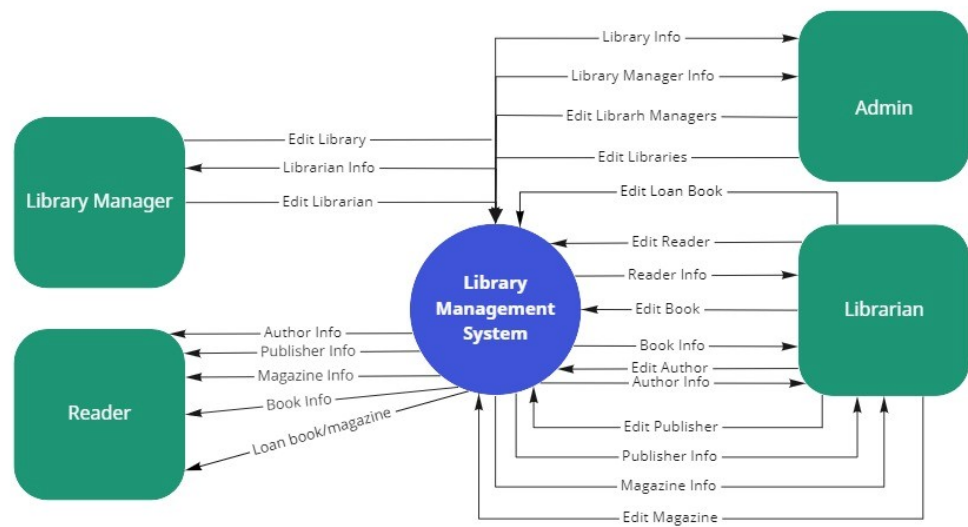
- We've used it to retain authors, publishers, libraries, and users. Thus, we used it to make it easy to search by name and id.

## 6 Use-Case Diagrams:



## 7 C4 Model of the System:

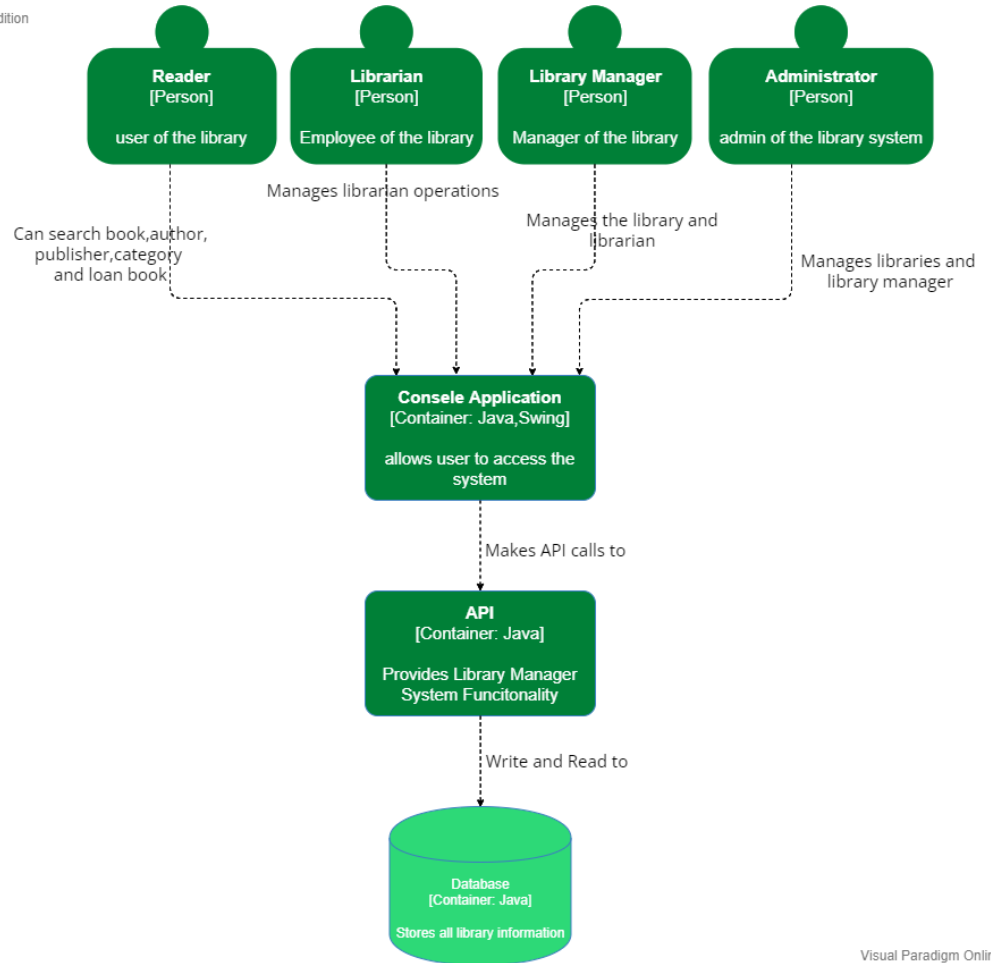
### 7.1 Context Model:





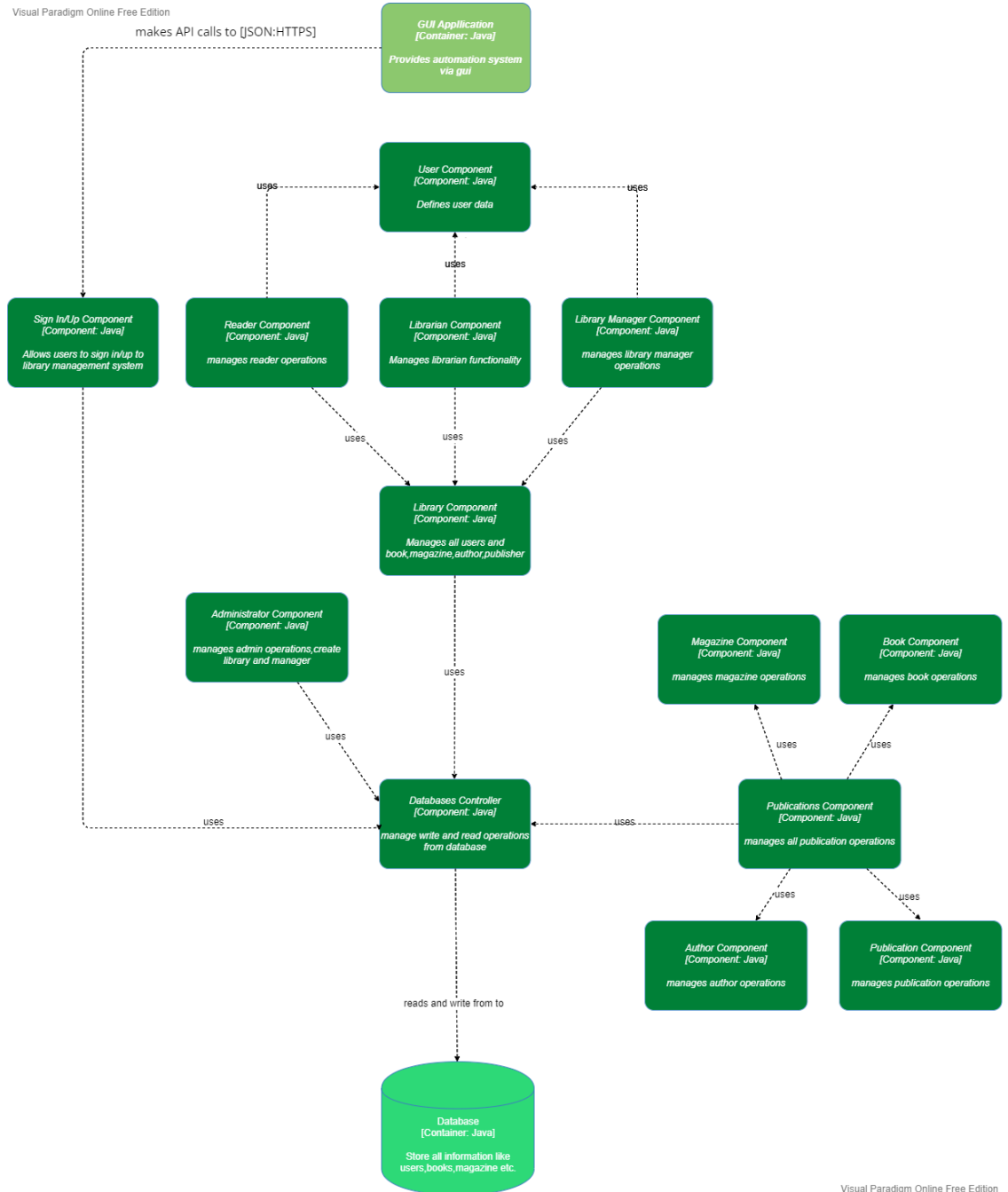
## 7.2 Container Diagram:

Visual Paradigm Online Free Edition

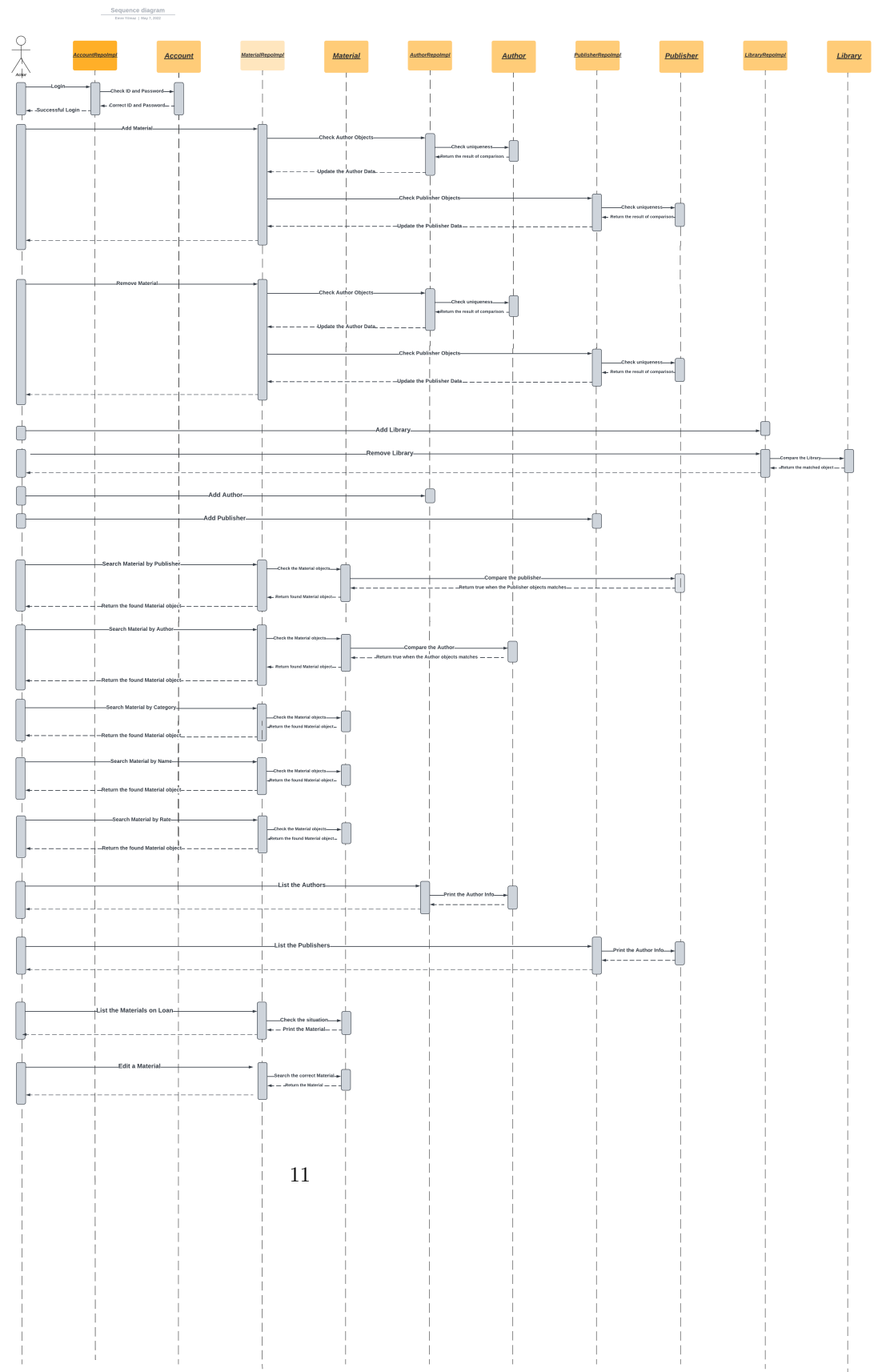


Visual Paradigm Online Free Edition

### 7.3 Component Diagram:

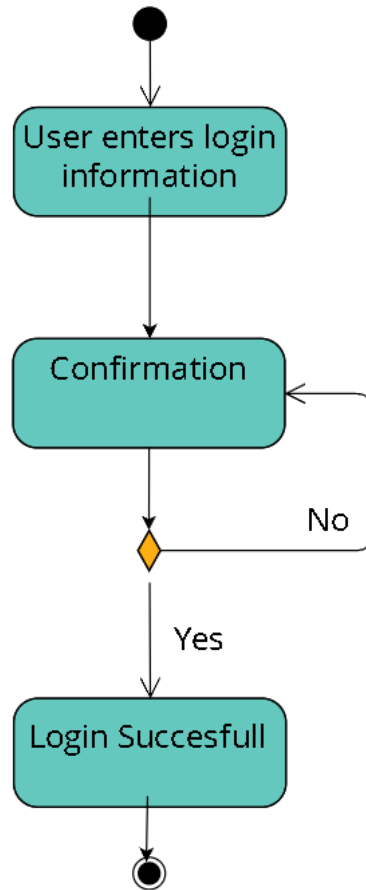


## 7.4 Sequence Diagram:

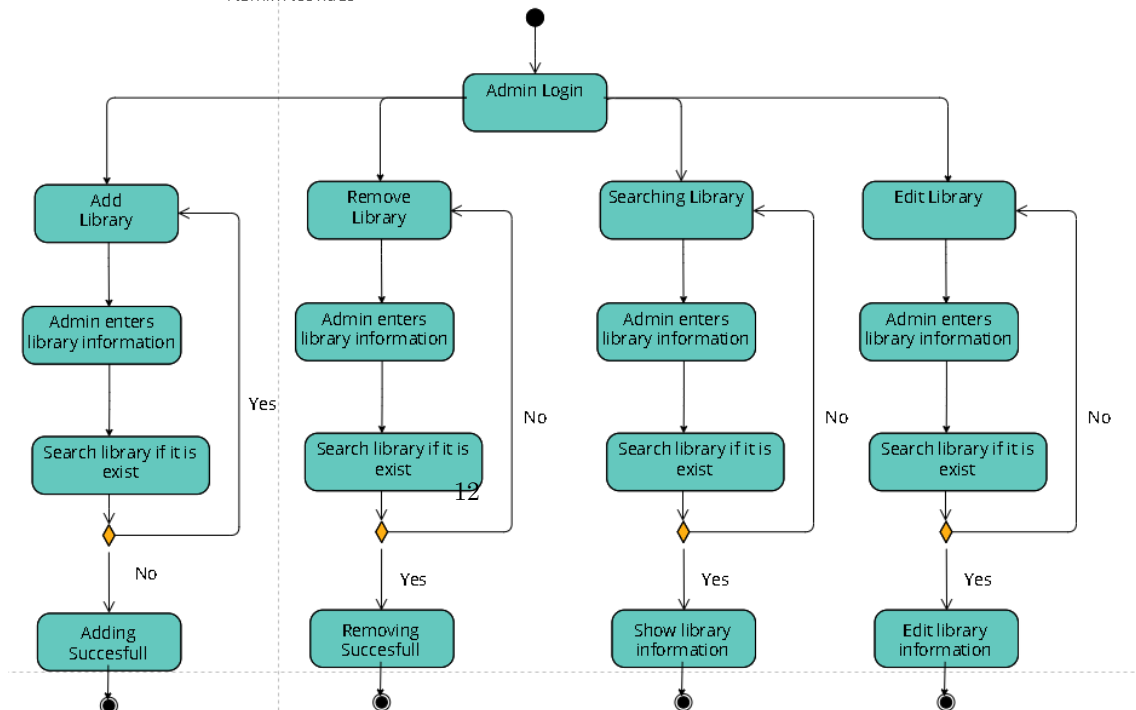


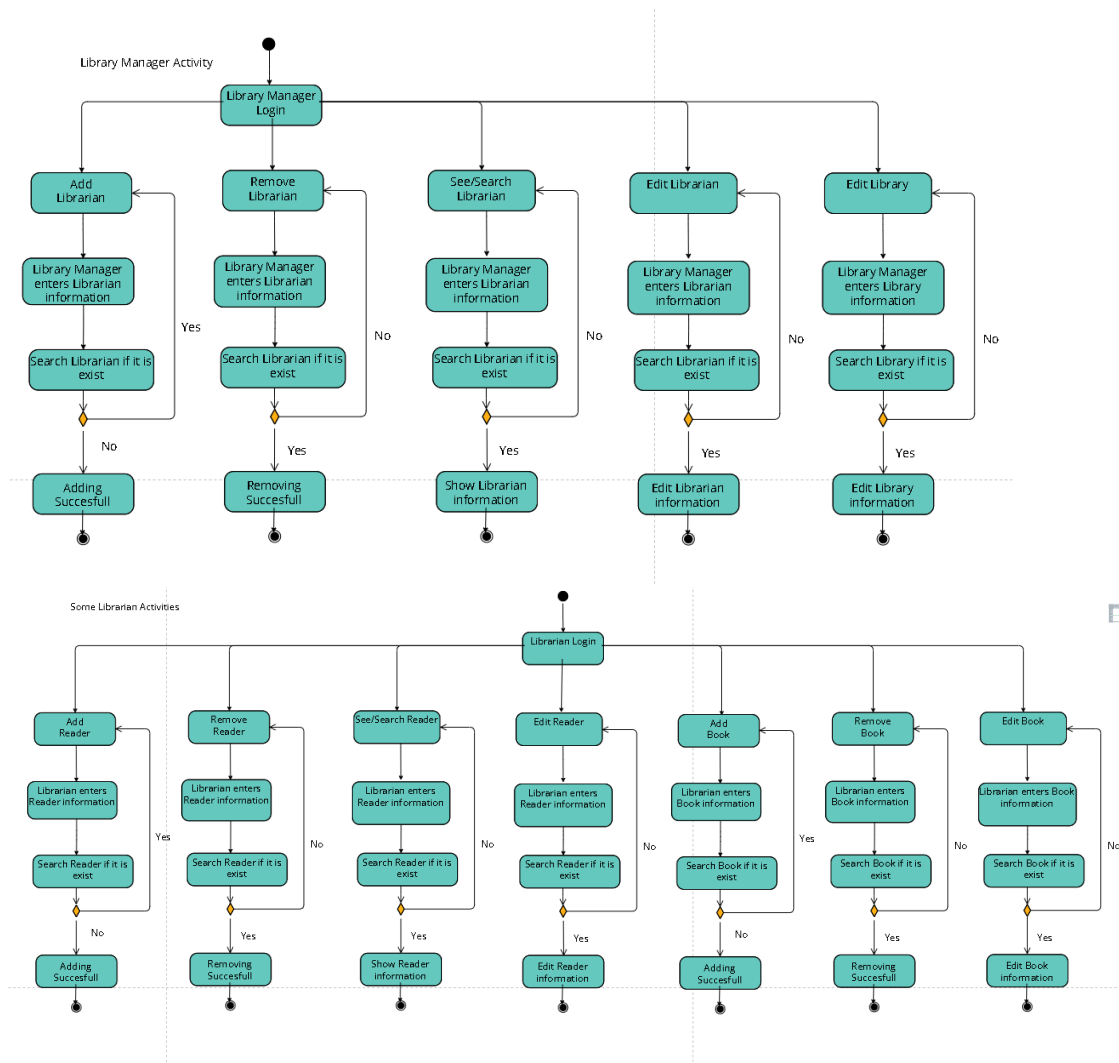
## 7.5 Activity Diagram:

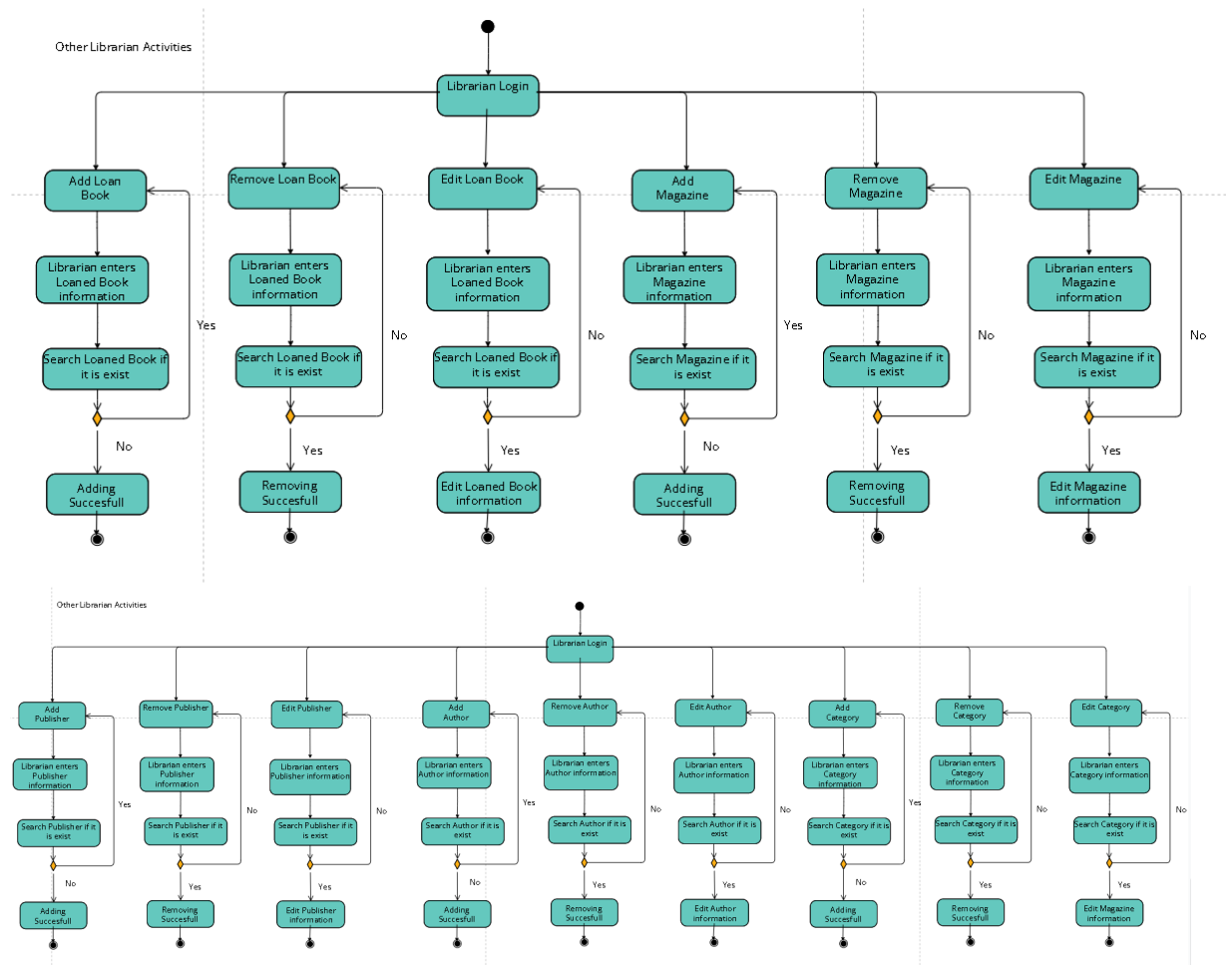
User Login Activity

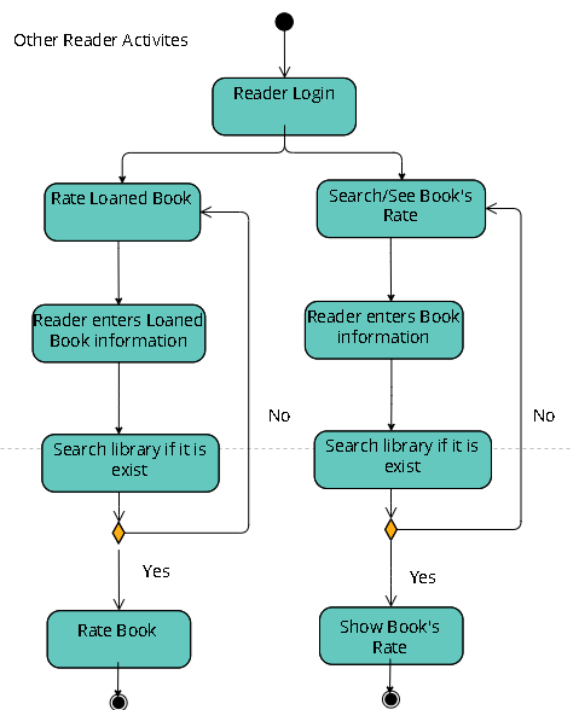
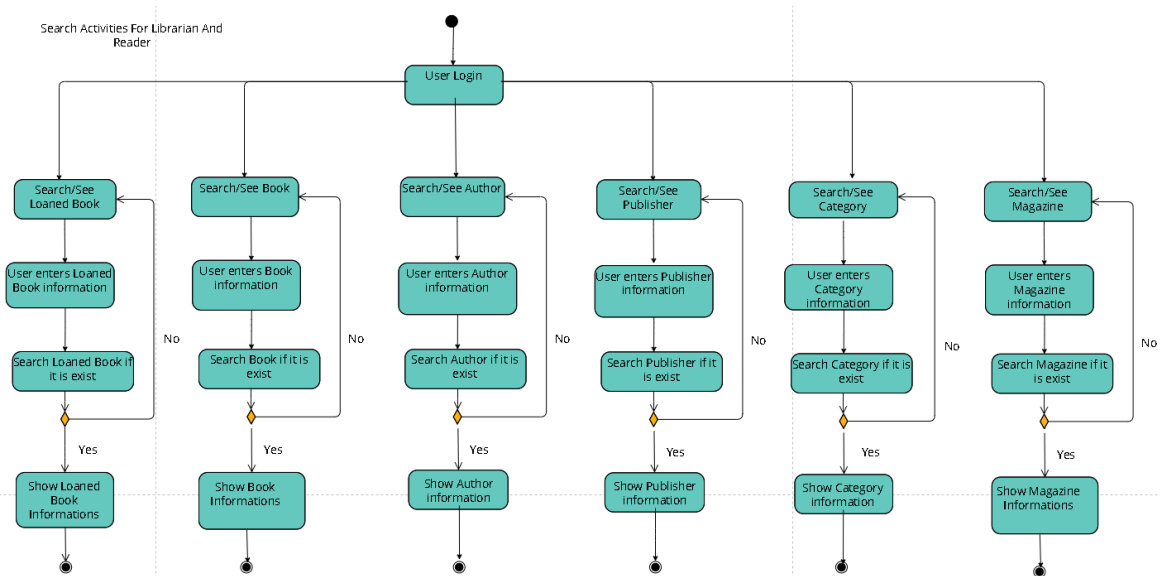


Admin Activities









## 7.6 Class Diagram:





## 8 Non-trivial Implementation Details:

### 8.1 Material:

- Material(Long, MaterialType, String, Category, Date, Author, Publisher, int, Situation, Location): We will use the given info to construct a Material Object.
- addRate(Integer): An user will give a rating for the material and this method pushes rate to the stack.

### 8.2 Author:

- Author(Long, String, String, String): We will use the given info to construct a Author Object.
- addBook(Material): Adds a new material to the list.
- removeBook(Material): Removes the given material from the list.

### 8.3 User:

- User(Long, String, String, String, Library): We will use the given info to construct a User Object.
- addMaterial(Material): Adds a new material to the list.
- searchMaterial(Material): Calls the contains() method of the list to see whether the material exist or not.
- removeMaterial(Material): Removes the given material from the list.

### 8.4 Publish:

- Publisher(Long, String, List<Material>, String): We will use the given info to construct a Publisher Object.
- addBook(Material): Adds a new material to the list.
- remove(Material): Removes the given material from the list.

### 8.5 PublisherRepositoryImpl:

- PublisherRepositoryImpl(): Default constructor to instantiate the object.
- remove(Publisher): Finds the given Publisher in BST. If exists, removes from the tree and rearranges the tree.
- findById(Long): Calls the find() method of BST with the given id.
- create(Publisher): Adds the given Publisher to the BST with the add() method of BST.
- findAll(): Traverses the tree and returns all the publishers in a list.
- update(Publisher): Finds the given Publisher by id and replaces it, returns the old data.

-viewInfo(Long): Finds the Publisher with the given id and if exists returns its info.

## 8.6 AuthorRepositoryImpl:

-AuthorRepositoryImpl(): Default constructor to instantiate the object.  
-remove(Author): Finds the given Author in BST. If exists, removes from the tree and rearranges the tree.  
-findById(Long): Calls the find() method of BST with the given id.  
-create(Author): Adds the given Author to the BST with the add() method of BST.  
-findAll(): Traverses the tree and returns all the authors in a list.  
-update(Author): Finds the given Author by id and replaces it, returns the old data.  
-viewInfo(Long): Finds the Author with the given id and if exists returns its info.

## 8.7 LibraryRepositoryImpl:

- LibraryRepositoryImpl(): Default constructor to instantiate the object.  
-remove(Library): Finds the given Library in the list. If exists, removes from the list.  
-findById(Long): Finds the library with the given id by doing linear search.  
-create(Library): Adds the given Library to the list with the add() method of list.  
-findAll():Returns all the libraries in the list.  
-update(Library): Finds the given Library by id and replaces it, returns the old data.  
-viewInfo(Long): Finds the Library with the given id and if exists returns its info.

## 8.8 MaterialRepositoryImpl:

- MaterialRepositoryImpl(): Default constructor to instantiate the object.  
-remove(Material): Finds the given Material in the queue. If exists, removes from the queue.  
-findById(Long): Finds the Material with the given id by calling peek() method of queue.  
-create(Material): Adds the given Material to the queue. -findAll():Returns all the materials in the queue.  
-update(Material): Finds the given Material by id and replaces it, returns the old data.  
-viewInfo(Long): Finds Library with the given id and if exists returns its info.

## 8.9 AccountRepositoryImpl:

- AccountRepositoryImpl(): Default constructor to instantiate the object.
- remove(Account): Finds the given Account in BST. If exists, removes from the tree and rearranges the tree.
- findById(Long): Calls the find() method of BST with the given id.
- create(Account): Adds the given Account to the BST with the add() method of BST.
- findAll(): Traverses the tree and returns all the authors in a list.
- update(Account): Finds the given Account by id and replaces it, returns the old data.
- viewInfo(Long): Finds the Account with the given id and if exists returns its info.

## 9 Performance Analysis:

| Method / Data Type         | Time Complexity | Experimental results |           |            |
|----------------------------|-----------------|----------------------|-----------|------------|
|                            |                 | 100 data             | 1000 data | 10000 data |
| <b>Material class</b>      | (1)             |                      |           |            |
| setters and getters        | (1)             |                      |           |            |
| addRate / Stack            | (1)             | 234 ms               | 262 ms    | 253 ms     |
| removeRate / Stack         | (1)             |                      |           |            |
| compareTo                  | (1)             |                      |           |            |
| equals                     | (1)             |                      |           |            |
| toString                   | (1)             |                      |           |            |
| <b>Author class</b>        |                 |                      |           |            |
| setters and getters        | (1)             |                      |           |            |
| addBook / ArrayList        | O(n)            | 293 ms               | 2823 ms   | 29837 ms   |
| removeBook / ArrayList     | O(n)            | 279 ms               | 2937 ms   | 32749 ms   |
| compareTo                  | (1)             |                      |           |            |
| equals                     | (1)             |                      |           |            |
| <b>Account class</b>       |                 |                      |           |            |
| setters and getters        | (1)             |                      |           |            |
| equals                     | (1)             |                      |           |            |
| compareTo                  | (1)             |                      |           |            |
| hashCode                   | (1)             |                      |           |            |
| <b>User class</b>          |                 |                      |           |            |
| addMaterial / LinkedList   | O(n)            | 361 ms               | 3721 ms   | 35573 ms   |
| searchMaterial /LinkedList | O(n)            | 412 ms               | 3982 ms   | 39921 ms   |
| removeMaterial /LinkedList | O(n)            | 372 ms               | 3895 ms   | 37459 ms   |
| <b>Library class</b>       |                 |                      |           |            |
| equals                     | (1)             |                      |           |            |
| hashCode                   | (1)             |                      |           |            |
| compareTo                  | (1)             |                      |           |            |
| <b>Publisher class</b>     |                 |                      |           |            |
| setters and getters        | (1)             |                      |           |            |
| addBook / ArrayList        | O(n)            | 285 ms               | 3243 ms   | 30257 ms   |
| remove / ArrayList         | O(n)            | 304 ms               | 2935 ms   | 29384 ms   |
| compareTo                  | (1)             |                      |           |            |
| equals                     | (1)             |                      |           |            |
| hashCode                   | (1)             |                      |           |            |

## 10 Test Cases:

| Test Case Id                          | Test Case Objective | Pre Requisite                                                                       | Steps                                                         | Input Data                             | Expect Output                      | Status |
|---------------------------------------|---------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------|----------------------------------------|------------------------------------|--------|
| <b>Test Cases For Administrator</b>   |                     |                                                                                     |                                                               |                                        |                                    |        |
| Admin_TC_01                           | Login               | Admin account register or create before                                             | 1. Enter User Name<br>2. Enter Password                       | 1. Admin<br>2. Pass                    | Loggin in                          | PASS   |
| Admin_TC_02                           | Login               | Admin account register or create before                                             | 1. Enter User Name<br>2. Enter Password                       | 1. Admin<br>2. wrongPass               | Wrong password or UserName         | FAIL   |
| Admin_TC_03                           | Remove Library      | There should be a library created with this id                                      | 1. Enter Library Id                                           | 1. 1001                                | Library removed                    | PASS   |
| Admin_TC_04                           | Remove Library      | There should be a library created with this id                                      | 1. Enter Library Id                                           | 1. 9999                                | There is no library with this id   | FAIL   |
| Admin_TC_05                           | Edit Library        | There should be a library created with this id                                      | 1. Enter Library Id                                           | 1. 1001                                | Library edited                     | PASS   |
| Admin_TC_06                           | Edit Library        | There should be a library created with this id                                      | 1. Enter Library Id                                           | 1. 9999                                | There is no library with this id   | FAIL   |
| <b>Test Cases For Library Manager</b> |                     |                                                                                     |                                                               |                                        |                                    |        |
| Manager_TC_01                         | Login               | Library Manager account registired or create before                                 | 1. Enter User Name<br>2. Enter Password                       | 1. Mngr1<br>2. 123Ab                   | Loggin in                          | PASS   |
| Manager_TC_02                         | Login               | Library Manager account registired or create before                                 | 1. Enter User Name<br>2. Enter Password                       | 1. wrongUserName<br>2. 123Ab           | Wrong password or UserName         | FAIL   |
| Manager_TC_03                         | Edit Library        | There should be a library created with this id                                      | 1. Enter Library Id                                           | 1. 1001                                | Library edited                     | PASS   |
| Manager_TC_04                         | Edit Library        | There should be a library created with this id                                      | 1. Enter Library Id                                           | 1. 9999                                | There is no library with this id   | FAIL   |
| Manager_TC_05                         | Add Librarian       | Librarian UserName shouldn't be used before and all required field should be filled | 1. Enter User Name<br>2. Enter Password<br>3. Enter LibraryNo | 1. newLibrarian<br>2. 123Ab<br>3. 1001 | Librarian added                    | PASS   |
| Manager_TC_06                         | Add Librarian       | Librarian UserName shouldn't be used before and all required field should be filled | 1. Enter User Name<br>2. Enter Password<br>3. Enter LibraryNo | 1. Mngr1<br>2. 123Ab<br>3. 1001        | User name already used             | FAIL   |
| Manager_TC_07                         | Add Librarian       | Librarian UserName shouldn't be used before and all required field should be filled | 1. Enter User Name<br>2. Enter Password<br>3. Enter LibraryNo | 1. newLibrarian<br>2. 123Ab<br>3. 9999 | There is no library with this id   | FAIL   |
| Manager_TC_08                         | Remove Librarian    | There should be a librarian created with this id                                    | 1. Enter Librarian Id                                         | 1. 1001                                | Librarian removed                  | PASS   |
| Manager_TC_09                         | Remove Librarian    | There should be a librarian created with this id                                    | 1. Enter Librarian Id                                         | 1. 9999                                | There is no librarian with this id | FAIL   |
| Manager_TC_10                         | Edit Librarian      | There should be a librarian created with this id                                    | 1. Enter librarian Id                                         | 1. 1001                                | Librarian edited                   | PASS   |
| Manager_TC_11                         | Edit Librarian      | There should be a librarian created with this id                                    | 1. Enter librarian Id                                         | 1. 9999                                | There is no librarian with this id | FAIL   |
| Manager_TC_12                         | Search Librarian    | No Pre Request                                                                      | 1. Enter librarian Id                                         | 1. 1001                                | Librarian Found                    | PASS   |
| <b>Test Cases For Reader</b>          |                     |                                                                                     |                                                               |                                        |                                    |        |
| Reader_TC_01                          | Login               | Reader account register or create before                                            | 1. Enter User Name<br>2. Enter Password                       | 1. Reader1<br>2. thisReader            | Loggin in                          | PASS   |
| Reader_TC_01                          | Login               | Reader account register or create before                                            | 1. Enter User Name<br>2. Enter Password                       | 1. wrongUserName<br>2. thisReader      | Wrong password or UserName         | FAIL   |
| Reader_TC_03                          | Rate Book           | The book should be already created                                                  | 1. Enter Book Id<br>2. Enter rate                             | 1. 1001<br>2. 4.5                      | Book rated                         | PASS   |
| Reader_TC_03                          | Rate Book           | The book should be already created                                                  | 1. Enter Book Id<br>2. Enter rate                             | 1. 9999<br>2. 4.5                      | There is no book with this id      | PASS   |
| Reader_TC_04                          | Search Loan book    | No Pre Request                                                                      | 1. Enter Load book id                                         | 1. 1001                                | Load Book searched with id         | PASS   |
| Reader_TC_05                          | Search Book         | No Pre Request                                                                      | 1. Enter Book Id                                              | 1. 1001                                | Book searched with id              | PASS   |
| Reader_TC_06                          | Search Author       | No Pre Request                                                                      | 1. Enter Author Id                                            | 1. 1001                                | Author searched with id            | PASS   |
| Reader_TC_07                          | Search Publisher    | No Pre Request                                                                      | 1. Enter Publisher Id                                         | 1. 1001                                | Publisher searched with id         | PASS   |
| Reader_TC_08                          | Search Category     | No Pre Request                                                                      | 1. Enter Category Id                                          | 1. 1001                                | Category searched with id          | PASS   |

| Test Cases For Librarian |                 |                                                                                   |                                                                                                                                                                                                  |                                                                                                             |                                                 |      |
|--------------------------|-----------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------|------|
| LibrarianTC_01           | Login           | Librarian account register or create before                                       | 1. Enter User Name<br>2. Enter Password                                                                                                                                                          | 1. Librarian1<br>2. thisLibrarian                                                                           | Loggin in                                       | PASS |
| LibrarianTC_02           | Login           | Librarian account register or create before                                       | 1. Enter User Name<br>2. Enter Password                                                                                                                                                          | 1. wrongUserName<br>2. thisLibrarian                                                                        | Wrong password or UserName                      | FAIL |
| LibrarianTC_03           | Add Reader      | Reader UserName shouldn't be used before and all required field should be filled. | 1. Enter User Name<br>2. Enter Password<br>3. Enter LibraryNo                                                                                                                                    | 1. newReader<br>2. 123Ab<br>3. 1001                                                                         | Reader added                                    | PASS |
| LibrarianTC_04           | Add Reader      | Reader UserName shouldn't be used before and all required field should be filled. | 1. Enter User Name<br>2. Enter Password<br>3. Enter LibraryNo                                                                                                                                    | 1. oldReader<br>2. 123Ab<br>3. 1001                                                                         | User name already used                          | FAIL |
| LibrarianTC_05           | Add Reader      | Reader UserName shouldn't be used before and all required field should be filled. | 1. Enter User Name<br>2. Enter Password<br>3. Enter LibraryNo                                                                                                                                    | 1. newLibrarian<br>2. 123Ab<br>3. 9999                                                                      | There is no library with this id                | FAIL |
| LibrarianTC_06           | Remove Reader   | There should be a reader created with this id                                     | 1. Enter Reader Id                                                                                                                                                                               | 1. 1001                                                                                                     | Reader removed                                  | PASS |
| LibrarianTC_07           | Remove Reader   | There should be a reader created with this id                                     | 1. Enter Reader Id                                                                                                                                                                               | 1. 9999                                                                                                     | There is no reader with this id                 | FAIL |
| LibrarianTC_08           | Edit Reader     | There should be a reader created with this id                                     | 1. Enter Reader Id                                                                                                                                                                               | 1. 1001                                                                                                     | Reader edited                                   | PASS |
| LibrarianTC_09           | Edit Reader     | There should be a reader created with this id                                     | 1. Enter Reader Id                                                                                                                                                                               | 1. 9999                                                                                                     | There is no reader with this id                 | FAIL |
| LibrarianTC_10           | Search Reader   | No Pre Request                                                                    | 1. Enter Load book id                                                                                                                                                                            | 1. 1001                                                                                                     | Reader searched with id                         | PASS |
| LibrarianTC_11           | Add Metarial    | Metarial id shouldn't be used before and all required field should be filled.     | 1. Enter Id<br>2. Enter Name<br>3. Enter CategoryId<br>4. Enter PublisherId<br>5. Enter Stuation<br>6. Enter PageCount<br>7. Enter PublicationDate<br>8. Enter Location<br>9. Enter MetarialType | 1. 1001<br>2. Sefiller<br>3. 1001<br>4. 1001<br>5. STORAGE<br>6. 200<br>7. 11.04.2001<br>8. C3<br>9. Book   | Book added                                      | PASS |
| LibrarianTC_12           | Add Metarial    | Metarial id shouldn't be used before and all required field should be filled.     | 1. Enter Id<br>2. Enter Name<br>3. Enter CategoryId<br>4. Enter PublisherId<br>5. Enter Stuation<br>6. Enter PageCount<br>7. Enter PublicationDate<br>8. Enter Location<br>9. Enter MetarialType | 1. 1002<br>2. Science<br>3. 1001<br>4. 1001<br>5. LIBRARY<br>6. 30<br>7. 06.03.2022<br>8. A2<br>9. Magazine | Magazine added                                  | PASS |
| LibrarianTC_13           | Add Metarial    | Metarial id shouldn't be used before and all required field should be filled.     | 1. Enter Id<br>2. Enter Name<br>3. Enter CategoryId<br>4. Enter PublisherId<br>5. Enter Stuation<br>6. Enter PageCount<br>7. Enter PublicationDate<br>8. Enter Location<br>9. Enter MetarialType | 1. 1001<br>2. Sefiller<br>3. 1001<br>4. 1001<br>5. STORAGE<br>6. 200<br>7. 11.04.2001<br>8. C3<br>9. Book   | The metarial id is already using                | FAIL |
| LibrarianTC_14           | Add Metarial    | Metarial id shouldn't be used before and all required field should be filled.     | 1. Enter Id<br>2. Enter Name<br>3. Enter CategoryId<br>4. Enter PublisherId<br>5. Enter Stuation<br>6. Enter PageCount<br>7. Enter PublicationDate<br>8. Enter Location<br>9. Enter MetarialType | 1. 1001<br>2. Sefiller<br>3. 1001<br>4. 1001<br>5. STORAGE<br>6. 200<br>7. 11.04.2001<br>8. C3<br>9. Book   | There is no Category and publisher with this id | FAIL |
| LibrarianTC_15           | Remove Metarial | There should be a metarial created with this id                                   | 1. Enter metarial Id                                                                                                                                                                             | 1. 1001                                                                                                     | Metarial removed                                | PASS |

|                |                  |                                                 |                                                                                   |                                                                                                 |                                          |      |
|----------------|------------------|-------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|------------------------------------------|------|
| LibrarianTC_16 | Remove Metarial  | There should be a metarial created with this id | 1. Enter metarial Id                                                              | 1. 9999                                                                                         | There is no metarial with this id        | FAIL |
| LibrarianTC_17 | Edit Metarial    | There should be a metarial created with this id | 1. Enter metarial Id                                                              | 1. 1001                                                                                         | Metarial edited                          | PASS |
| LibrarianTC_18 | Edit Metarial    | There should be a metarial created with this id | 1. Enter metarial Id                                                              | 1. 9999                                                                                         | There is no metarial with this id        | FAIL |
| LibrarianTC_19 | Search Metarial  | No Pre Request                                  | 1. Enter metarial Id                                                              | 1. 1001                                                                                         | Book searched with id                    | PASS |
| LibrarianTC_20 | Add Loan Book    | The book should be on library.                  | 1. Enter Metarial Id                                                              | 1. 1001                                                                                         | Metarial given user login in             | PASS |
| LibrarianTC_21 | Add Loan Book    | The book should be on library.                  | 1. Enter Metarial Id                                                              | 1. 9999                                                                                         | There is no metarial with this id        | FAIL |
| LibrarianTC_22 | Add Loan Book    | The book should be on library.                  | 1. Enter Metarial Id                                                              | 1. 1002                                                                                         | The Metarial is already in diffrent user | FAIL |
| LibrarianTC_23 | Edit Loan Book   | The book should be on library.                  | 1. Enter metarial Id<br>2. Enter metarial stuation                                | 1. 1001<br>2. LIBRARY                                                                           | Metarial Stuation is changed             | PASS |
| LibrarianTC_24 | Edit Loan Book   | The book should be on library.                  | 1. Enter metarial Id<br>2. Enter metarial stuation                                | 1. 9999<br>2. LIBRARY                                                                           | There is no metarial with this id        | FAIL |
| LibrarianTC_25 | Search Loan book | No Pre Request                                  | 1. Enter Load book id                                                             | 1. 1001                                                                                         | Loan Books searched with id              | PASS |
| LibrarianTC_26 | Add Publisher    | The Published id shouldn't be used before       | 1. Enter Publisher Id<br>2. Enter Name<br>3. Enter Info<br>4. Enter Metarial List | 1. 1001<br>2. KVK kitabevi<br>3. KVK kitabevi 1978 senesinde Istanbul...<br>4. Satranç Sefiller | Publisher is added                       | PASS |
| LibrarianTC_27 | Add Publisher    | The Published id shouldn't be used before       | 1. Enter Publisher Id<br>2. Enter Name<br>3. Enter Info<br>4. Enter Metarial List | 1. 9999<br>2. Stefan Zweig<br>3. Stefan Zweig is born in ...<br>4. Satranç Amok Koşucusu        | Publisher is already in system           | FAIL |
| LibrarianTC_28 | Search Publisher | No Pre Request                                  | 1. Enter Publisher Id                                                             | 1. 1001                                                                                         | Publisher searched with id               | PASS |
| LibrarianTC_29 | Add Author       | The Author id shouldn't be used before          | 1. Enter Author Id<br>2. Enter Name<br>3. Enter Info<br>4. Enter Metarial List    | 1. 1001<br>2. Stefan Zweig<br>3. Stefan Zweig is born in ...<br>4. Satranç Amok Koşucusu        | Author is added                          | PASS |
| LibrarianTC_30 | Add Author       | The Author id shouldn't be used before          | 1. Enter Author Id<br>2. Enter Name<br>3. Enter Info<br>4. Enter Metarial List    | 1. 9999<br>2. Stefan Zweig<br>3. Stefan Zweig is born in ...<br>4. Satranç Amok Koşucusu        | Author is already in system              | FAIL |
| LibrarianTC_31 | Search Author    | No Pre Request                                  | 1. Enter Author Id                                                                | 1. 1001                                                                                         | Author searched with id                  | PASS |
| LibrarianTC_32 | Search Category  | No Pre Request                                  | 1. Enter Category Id                                                              | 1. 1001                                                                                         | Category searched with id                | PASS |

## 11 Result of Unit Tests:

The screenshot displays the IntelliJ IDEA test runner interface, showing the results of four unit tests. Each test is expanded to show its individual methods and their execution times. The tests are: testAdministrator, testLibraryManager, testReader, and testLibrarian. The testReader and testLibrarian tests include several disabled tests marked with a grey circle icon.

```
Run: testAdministrator x
  ✓ testAdministrator (librarymanagement 183 ms) /home/karacete/.jdk/openjdk-18.0.1.1/bin/java ...
    ✓ testLogin() 176 ms
    ✓ testRemoveLibrary() 4 ms
    ✓ testEditLibrary() 3 ms
    Process finished with exit code 0

  ✓ testLibraryManager (librarymanagemen 46 ms) /home/karacete/.jdk/openjdk-18.0.1.1/bin/java ...
    ✓ removeLibrarian() 34 ms
    ✓ testLogin() 3 ms
    ✓ testEditLibrarian() 3 ms
    ✓ addLibrarian() 2 ms
    ✓ testEditLibrary() 4 ms
    Process finished with exit code 0

  ✗ testReader (librarymanagementsystem. 147 ms) /home/karacete/.jdk/openjdk-18.0.1.1/bin/java ...
    ✓ testSearchLoanBook() 36 ms
    ✓ testSearchBook() 3 ms
    ✓ testLogin() 3 ms
    ✓ testSearchPublisher() 3 ms
    ✗ testSearchByCategory() 1 ms void librarymanagementsystem.test.testReader.testSearchByCategory() is @Disabled
    ✓ testSearchAuthor() 1 ms
    ✗ testRateBook() 1 ms void librarymanagementsystem.test.testReader.testRateBook() is @Disabled
    Process finished with exit code 0

Run: testLibrarian x
  ✗ Tests passed: 9, ignored: 5 of 14 tests - 58 ms

  ✗ testLibrarian (librarymanagementsystem 58 ms) /home/karacete/.jdk/openjdk-18.0.1.1/bin/java ...
    ✓ testAddPublisher() 41 ms
    ✓ testAddMaterial() 4 ms void librarymanagementsystem.test.testLibrarian.testEditMaterial() is @Disabled
    ✓ testLogin() 2 ms
    ✓ testSearchPublisher() 3 ms void librarymanagementsystem.test.testLibrarian.testAddReader() is @Disabled
    ✗ testEditMaterial() 3 ms
    ✓ testRemoveReader() 2 ms void librarymanagementsystem.test.testLibrarian.testSearchCategory() is @Disabled
    ✓ testAddAuthor() 1 ms
    ✗ testAddReader() 1 ms void librarymanagementsystem.test.testLibrarian.testEditReader() is @Disabled
    ✓ testSearchAuthor() 1 ms
    ✗ testSearchCategory() 2 ms void librarymanagementsystem.test.testLibrarian.testRemoveMaterial() is @Disabled
    ✗ testEditReader() 2 ms
    ✓ testSearchMaterial() 2 ms
    ✓ testSearchLoanedBook() 2 ms
    ✗ testRemoveMaterial() 2 ms
    Process finished with exit code 0
```